DATAGENT: Data Analytics Toolkit for AutomaticGeneration of Dashboards and Tailored Insights

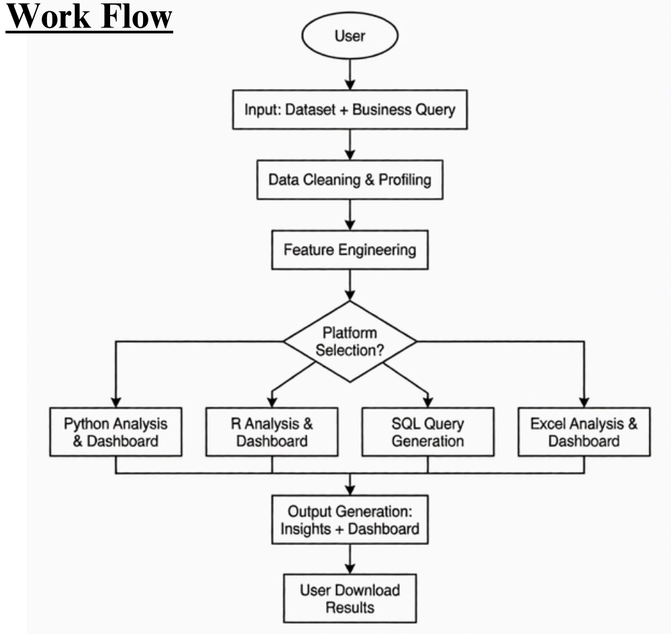
Problem Statement

Most existing data analytics workflows lack unified AI-driven automation for data cleaning, feature selection, multi-platform script generation, and dashboard creation, making the process time-consuming, complex, and less accessible for non-technical users.

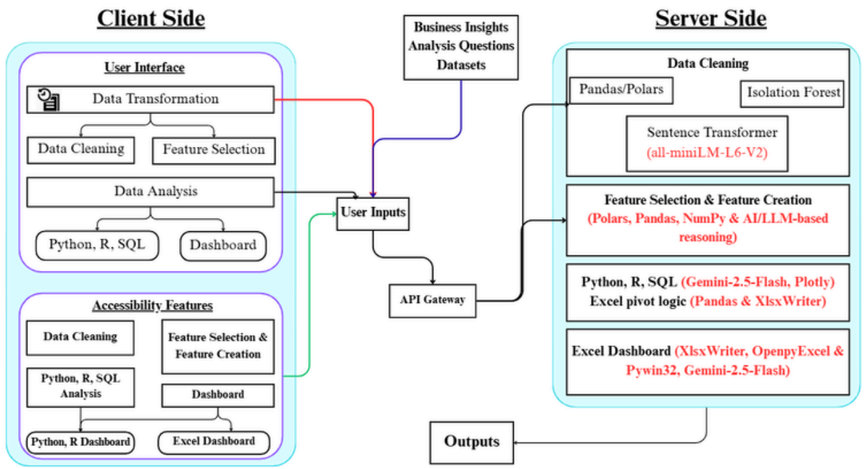
Objectives

- To automate data cleaning and imputation, using Polars, Pandas, Random Forest, Isolation Forest and KNN.
- To enable intelligent feature selection, leveraging NumPy, Polars, Pearson Corelation and LLM (Gemini).
- To generate analysis-ready scripts, using LLM for Python, R, SQL, and Excel logic.
- To create interactive and customizable dashboards, using Plotly, XlsxWriter, Openpyxl, and Pywin32.

Work Flow



System Architecture

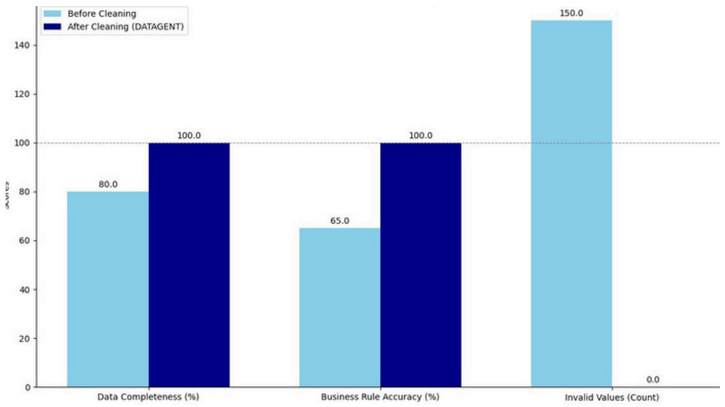


Python, R, Excel

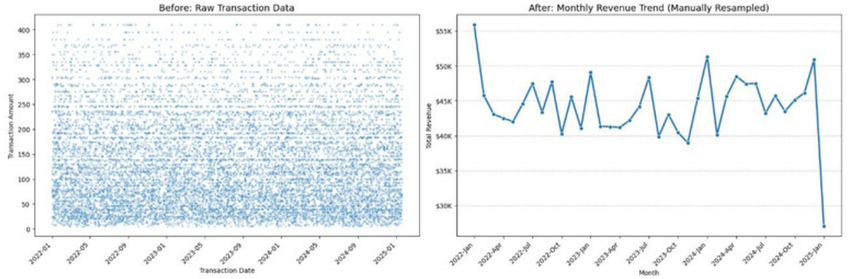
Analytical Question	Python (Pandas)	Excel Dashboard
Best Month for Sales	April ($28,444.25)	April ($28,444.25)
City with Highest Revenue	San Francisco ($8,211.39)	San Francisco ($8,211.39)
Most Frequent Product	USB-C Charging Cable (396)	USB-C Charging Cable (396)
Top 5 Sales Contribution	$14,946.94 (Combined)	$14,946.94 (Combined)

Results

Data Cleaning



Feature Engineering



Key References

[1] A. Mumuni and F. Mumuni, "Automated data processing and feature engineering for deep learning and big data applications: a survey," J. Inf. Intell., Jan. 2024, doi:10.1016/j.jiixd.2024.01.002.

[2] B. Praneeth, A. K. Singh, J. K. Raju, Mohana, and B. Suma, "Natural Language Interface for Data Visualization: Harnessing the Power of Large Language Models," in Proc. 15th Int. Conf. Comput. Commun. Netw. Technol. (ICCCNT), Kamand, India, 2024, pp. 1–6, doi: 10.1109/ICCCNT61001.2024.10724113.

[3] Y. Luo, X. Qin, N. Tang and G. Li, "DeepEye: Towards Automatic Data Visualization," 2018 IEEE 34th International Conference on Data Engineering (ICDE), Paris, France, 2018, pp. 101-112, doi: 10.1109/ICDE.2018.00019.

Conclusion

DATAGENT streamlines the entire data analytics process with AI-driven automation for data cleaning, feature selection, script generation, and dashboard creation, enabling faster, accessible, and efficient decision-making across users.